Worksheet 8.6

Science Quest 10: pages 316–17

Newton's Second Law

Answers

1. State Newton's Second Law of Motion.

The acceleration of a body is directly proportional to the net force and inversely proportional to its mass.

2. Use the formula: $F = ma$ to complete the following table

Force (F) (newtons)	Mass (m)	Acceleration (a)
100	40 kg	$\frac{100}{40} = 2.5\text{ m/s}^2$
2	200 g	$\frac{2}{0.20} = 10\text{ m/s}^2$
70	$\frac{70}{3.5} = 20\text{ kg}$	3.5 m/s^2
100	$\frac{100}{0.10} = 1000\text{ kg}$	10 cm/s^2

3. Consider the forces acting on a body of mass 40 kg resting on a frictionless surface.

(a) Calculate the net force acting on the body.

$F = 35 - 15 = 20\text{ N}$

(b) Calculate the acceleration of the body.

$a = \frac{F}{m} = \frac{20}{40} = 0.5\text{ m/s}^2$

4. The photo shows a drag car coming to a stop at the end of a race.

Photo: Claudia Schnepf / 123RF

(a) To come to a complete stop, the drag car negatively accelerates. Identify another term for 'negative acceleration'. Deceleration

(b) Identify the force that brings the drag car to a stop.

Friction